

Project Title: Cinema Database Management System**Team Members: Arzumanyan Natali, Grigoryan Zara, Mashinyan Srбуhi****Submission Phase: Phase 4 - Final Implementation**

Throughout the development of the Cinema Database Management System, our team gained comprehensive, hands-on experience across all stages of database design, implementation, and testing. We began by transforming real-world cinema operations into a structured conceptual model using ERD diagrams. This process required understanding specialization, weak entities, relationship cardinalities, and business rules such as preventing session overlaps and avoiding seat double-booking. We then mapped the conceptual model into a relational schema and implemented it fully using SQL. We learned how to design tables with correct primary and foreign keys, apply CHECK constraints for data validity, and enforce referential integrity using ON DELETE CASCADE. This phase strengthened our understanding of schema correctness, consistency, and production-level database design.

Throughout the project, we learned valuable technical and teamwork lessons:

- How to refine and evolve an ERD based on actual query needs
 - The importance of normalization for clarity and maintainability
 - How business rules translate into constraints, triggers, and query logic
 - The role of indexes in query performance and scheduling conflict detection
 - The value of realistic test data for validating system behavior
- Effective team communication and iterative problem-solving

Overall, the project helped us understand how a complete database system is designed, built, populated, and queried in practice. It strengthened our SQL, modeling, and analytical thinking skills and gave us real insight into how large-scale data systems support everyday operations such as ticketing, scheduling, payments, and reporting.